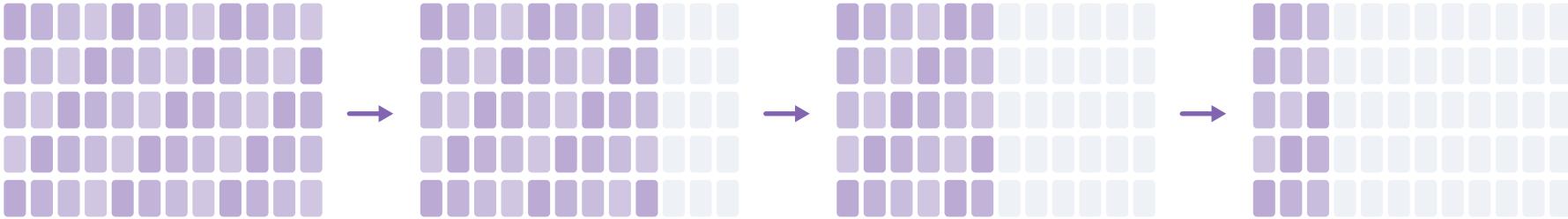
3 Choose descriptors + learner

Remove less useful predictors using training data.

Training-only importance for candidate ordering (TreeSHAP)



All columns **75%** **50%** **25%**

Columns retained from the original table

Refit after each reduction

Inner validation chooses the retained fraction and the model family.

Keep a feature budget + learner

Relearn the selected column mask before outer evaluation.